MA 374 (2026) Lab 11
Asian Option Pricing by Monte Carlo Simulation

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1 Introduction

In this lab, we study a stock price following a geometric Brownian motion (GBM) and use Monte Carlo simulation to price six-month Asian options with arithmetic averaging. We first simulate paths in both the real-world measure and the risk-neutral measure. We then compute Asian call and put prices for strikes $K = 105$, 110, and 90. Finally, we apply variance reduction techniques and compare the results.

2 Model Setup

The asset price is assumed to follow the GBM process

$$dS_t = \mu S_t \, dt + \sigma S_t \, dW_t$$

under the real-world measure, and

$$dS_t = r S_t \, dt + \sigma S_t \, dW_t$$

under the risk-neutral measure.

The parameter values used are:

$$S_0 = 100, \qquad \mu = 0.10, \qquad r = 0.05, \qquad \sigma = 0.20, \qquad T = 0.5.$$

For the Asian option, the payoff is based on the arithmetic average price:

$$\bar{S} = \frac{1}{n} \sum_{i=1}^{n} S_{t_i}.$$

Hence, the payoffs are

$$\text{Call payoff} = \max(\bar{S} - K, 0), \qquad \text{Put payoff} = \max(K - \bar{S}, 0).$$

The discounted Monte Carlo estimate is

$$V_0 \approx e^{-rT} \frac{1}{N} \sum_{j=1}^{N} \text{Payoff}^{(j)}.$$

3 Question 1: Simulation and Plain Monte Carlo Pricing

3.1 Simulated Paths

Figure shows the simulated GBM paths in the real-world and risk-neutral settings. As expected, the real-world paths have a slightly higher upward tendency because $\mu > r$.

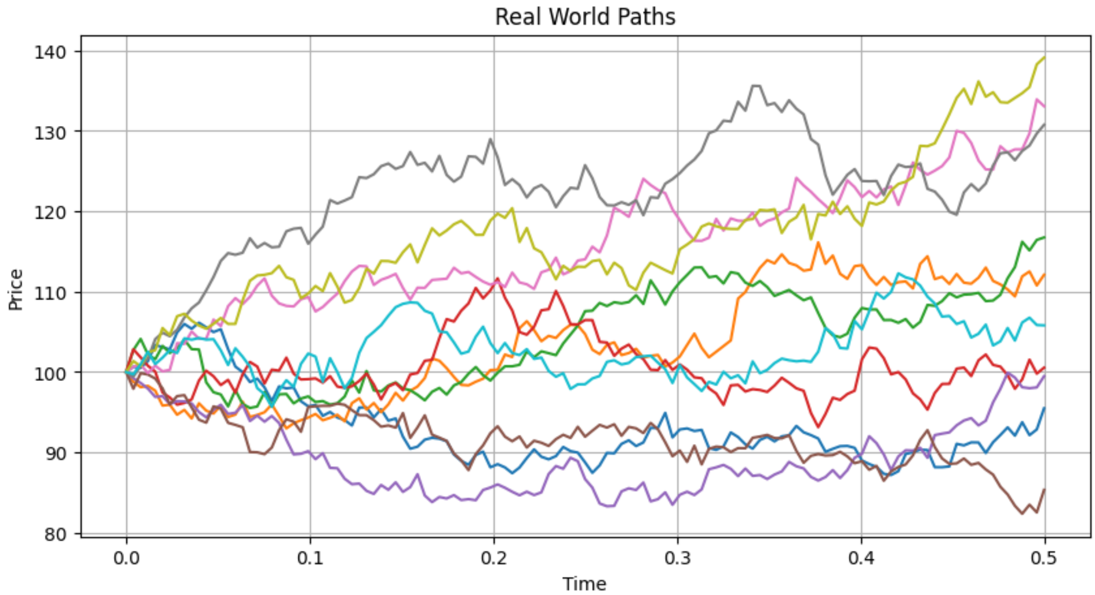

Figure 1: Simulated GBM paths under the real-world measure.

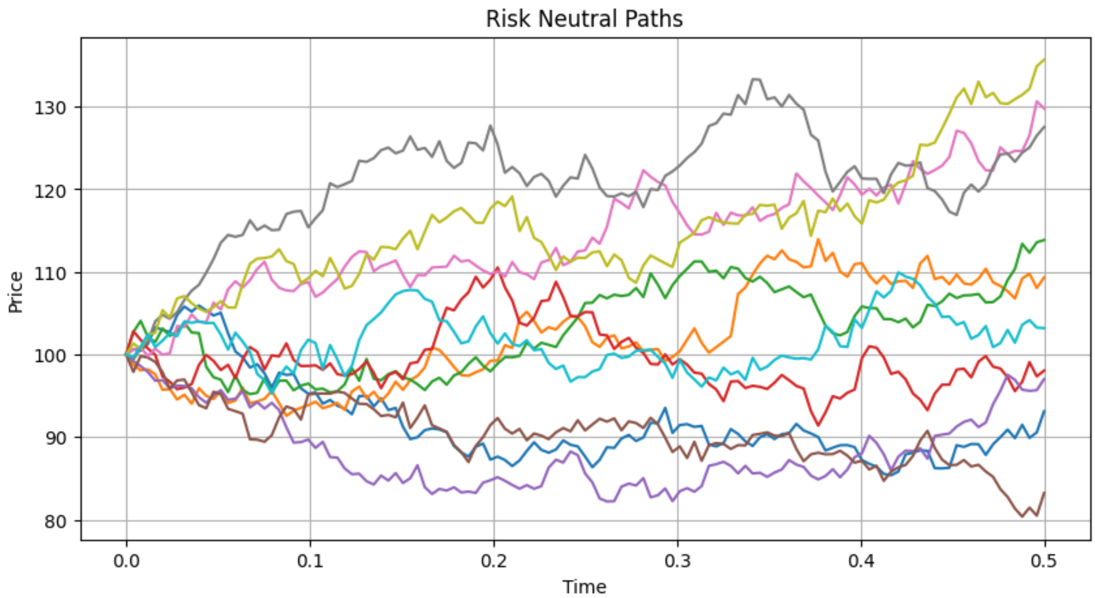

Figure 2: Simulated GBM paths under the risk-neutral measure.

3.2 Option Prices for Different Strikes

The Asian option prices obtained using plain Monte Carlo are shown below.

Table 1: Asian option prices from plain Monte Carlo simulation

Strike K	Call Price	Call 95% CI	Put Price	Put 95% CI
105	1.793435	[1.769834, 1.817035]	5.463614	[5.428378, 5.498849]
110	0.709297	[0.694541, 0.724054]	9.256026	[9.213257, 9.298795]
90	11.212748	[11.164952, 11.260544]	0.253278	[0.246296, 0.260261]

3.3 Strike Sensitivity

Figure 3 shows the variation of Asian option prices with strike price.

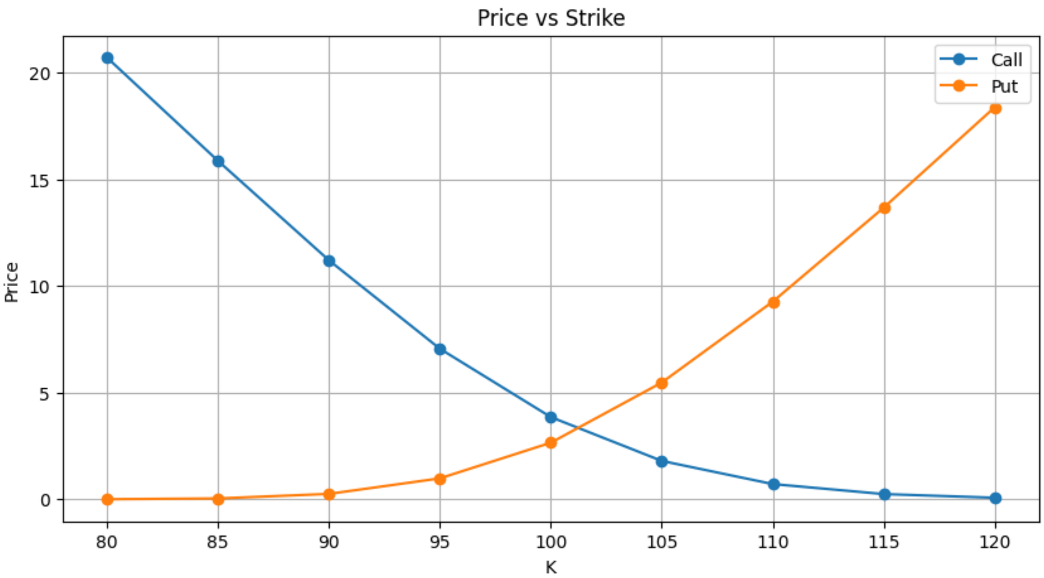

Figure 3: Asian option price versus strike price.

As expected, the call price decreases as the strike increases, while the put price increases.

3.4 Volatility Sensitivity

Figure 4 shows the effect of volatility on option prices.

Figure 4: Asian option price versus volatility.

The option prices increase with volatility because higher uncertainty increases the likelihood of large favorable payoffs.

4 Question 2: Variance Reduction

To improve Monte Carlo efficiency, we used two variance reduction methods:

- Antithetic variates
- Control variate method using the corresponding European option

4.1 Results for Variance Reduction

Table 2: Variance reduction comparison for Asian call and put options

Strike	Method	Call Price (SE)	Put Price (SE)
$K = 105$			
	Plain	1.842134 (0.017190)	5.426063 (0.025399)
	Antithetic	1.817848 (0.014978)	5.447061 (0.011553)
	Control variate	1.840261 (0.010125)	5.451278 (0.013911)
$K = 110$			
	Plain	0.734632 (0.010688)	9.195111 (0.030919)
	Antithetic	0.717164 (0.010092)	9.222927 (0.006725)
	Control variate	0.735776 (0.007080)	9.226589 (0.016582)
$K = 90$			
	Plain	11.294505 (0.034679)	0.248786 (0.005038)
	Antithetic	11.247761 (0.008213)	0.247325 (0.004837)
	Control variate	11.279138 (0.018002)	0.253412 (0.003687)

4.2 Discussion

The variance reduction methods preserve the option value while reducing the standard error. Antithetic variates consistently improved precision. The control variate method also reduced variance in most cases, especially when the Asian payoff and the European payoff were strongly correlated.

A key observation is that the standard error is much smaller under variance reduction than under plain Monte Carlo, which shows that the estimator is more efficient. In particular, the antithetic method performed very well for the put options with $K = 105$ and $K = 110$.

5 Conclusion

The Monte Carlo simulation produced sensible Asian option prices for all three strikes. The prices behaved as expected with respect to strike and volatility. Variance reduction significantly improved the efficiency of the simulation, with antithetic variates and control variates both reducing standard errors compared to plain Monte Carlo.